On the Conservation Laws and Pathway Realizability of Biochemical Reaction Networks

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Abstract

Background: Conserved metabolite pools and substrate availability constrain biochemical pathways before kinetic parameters are known. These properties are usually evaluated on incompatible mathematical abstractions: stoichiometric nullspaces omit shared catalytic participants, while graph reachability ignores token multiplicities. Because participants that appear identically on both reaction sides are eliminated under net stoichiometry despite being required for firing, standard reductions can destroy catalytic gating and alter deficiency.

Results: We present the chemical reaction network (CRN) submodule of `SynKit`, an open-source Python toolkit with an optimized C++ core. Networks imported from `KEGG`, structural `SBML` models, or generative rules share a bipartite representation preserving distinct reactant and product incidences. The toolkit computes exact conservation laws, minimal semiflows and siphons, deficiency, and discrete firing certificates for prescribed reaction multisets. Twelve reference networks match analytical values and agree with `CRNT4SBML` across eight structural invariants. Ten reaction-bearing `BioModels` entries, up to 504 species and 1 654 directed reactions, pass structural round trips. In one model, cancelling shared catalysts leaves net stoichiometry unchanged but lowers deficiency from 13 to 11. Matched formose expansions reach identical sets of 10 572 species and 11 239 reactions across six depths in `SynKit`, `MØD`, and `MolFoundry`. For a prescribed glycolytic flux with ADP, NAD^+ and phosphate supplied, one initial ATP gives a non-negative predicted final inventory yet blocks execution. Two initial ATP permit a 15-step certificate. In the 37-species formose network, conservation weights reproduce carbon counts. Two prescribed target routes require two or three glycolaldehyde molecules and consume no formaldehyde, explaining their different outcomes across inventories with equal carbon budgets.

Conclusions: Stoichiometric conservation is necessary but insufficient to guarantee pathway execution. By unifying conservation analysis, deficiency invariants, and Petri-net reachability on a single incidence representation with exact rational arithmetic, `SynKit` makes participant retention, cofactor boundaries, and precursor constraints explicit and testable. Source code, recorded benchmarks, and reproducible analysis workflows accompany the manuscript.

Keywords: Chemical reaction networks, chemical reaction network theory, Petri nets, conserved moieties, stoichiometry, pathway realizability, `SBML`

1 Background

Curated databases such as KEGG [1], Reactome [2], and WikiPathways [3] catalog known metabolic pathways and biochemical reactions, while graph-transformation methods enumerate candidate reactions from molecular seeds and transformation rules [4]. The rules used for such network generation can be extracted automatically from atom-mapped reaction records using SynTemp [5]. Before assigning kinetic parameters, structural analyses clarify the behavioral constraints imposed by network topology. Specifically, researchers examine which metabolite pools are conserved by stoichiometric invariants [6], which qualitative dynamical bounds or multistationarity criteria are governed by reaction network structure [7, 8], and whether a candidate pathway is realizable from a finite substrate inventory [9].

Assessing these constraints simultaneously is difficult because each analysis relies on a different mathematical abstraction of the biochemical system. A species graph captures connectivity between metabolites but loses the joint requirement of multiple co-substrates [10]. The net stoichiometric matrix records net changes in species abundance but omits participants that appear identically as reactants and products, such as catalysts or recycled cofactors. For example, a catalyzed reaction $A + C \to B + C$ and the uncatalyzed conversion $A \to B$ yield identical net columns in $\mathbf{S}$, yet only the first requires the catalyst C to proceed. Under mass-action kinetics its rate also depends on C. The standalone networks both have deficiency zero ($n = 2$, $\ell = 1$, $s = 1$), although their complexes differ. Within a larger network, cancelling shared participants can change complex identifications and linkage classes, and hence deficiency. The risk lies in tools that store reactions as net coefficient maps: COBRApy 0.32.1 [11], for example, stores $A + C \to B + C$ as $A \to B$, both when a reaction is built in Python and when it is read from SBML. Similarly, a balanced reaction combination or a qualitatively reachable product does not establish that the reactions can proceed in an admissible sequence from a finite inventory [9]. Representation choices during model import can therefore compromise the assumptions of another analysis.

Because of these distinct formulations, existing software packages specialize in separate subsets of analyses: COBRApy [11] supports constraint-based flux analysis [12], CRNT4SBML [13] evaluates deficiency and potential bistability, MØD [4] explores rule-generated reaction spaces, Tellurium [14] and libRoadRunner [15] simulate kinetic ODEs, and Petri-net methods analyze metabolic invariants [16, 17] and provide pathway realizability certificates [9]. Standard exchange formats such as SBML [18] and COMBINE [19], alongside converters like KEGGtranslator [20], facilitate model transfer. However, converting models between separate formats can inadvertently discard structural features, such as catalysts or boundary species, that specific analyses require.

Conserved pools and substrate sufficiency therefore need to be examined on the same underlying reaction model. In glycolysis [1], a route appears connected from glucose to pyruvate when cofactors are omitted, yet the prescribed flux cannot proceed when currency metabolites are retained and their initial inventories are zero. In a rule-generated formose network [21], every reaction may conserve carbon, yet a proposed pathway

remains unrealizable if its reactions require more glycolaldehyde molecules than the initial precursor pool provides. In both cases, conservation laws alone cannot establish whether a reaction sequence is realizable from a specified finite inventory [9].

Here, we present the chemical reaction network (CRN) submodule of `SynKit` [22] to evaluate these constraints on a shared representation. By connecting `KEGG` pathway import, structural `SBML` exchange, and rule-based generation to exact conservation laws, deficiency theory, and Petri-net reachability, the toolkit tests whether reaction sequences can execute without discarding catalysts or cofactors. We validate the implementation across twelve reference networks against `CRNT4SBML`, verify structural round trips on ten `BioModels` entries, and benchmark exact enumeration limits. A curated `BioModels` entry shows how participant retention changes deficiency without changing net stoichiometry. Case studies in glycolysis and formose show how cofactor boundaries and precursor inventories affect execution of prescribed reaction multisets.

2 Implementation

2.1 Network representation and participant incidence

`SynCRN` serves as the core reaction-network data structure in `SynKit`, maintaining species, reactions, transformation rules, and chemical annotations (Fig. 1). It derives bipartite digraphs in `NetworkX` [23], stoichiometric matrices, and Petri nets for downstream structural analysis. The bipartite digraph has species vertices X and reaction vertices $\mathscr{R}$, and each reaction vertex is linked to its reactants and products. It is therefore the incidence representation of a directed hypergraph in which every reaction is a hyperedge, so multi-substrate requirements remain joint hyperedge tails rather than pairwise species edges.

For reaction r, reactant and product multisets have coefficients s^-_{xr} and s^+_{xr}, defining separate reactant and product (pre/post) incidence matrices $\mathbf{S}^- = (s^-_{xr})$ and $\mathbf{S}^+ = (s^+_{xr})$ alongside the net stoichiometry $\mathbf{S} = \mathbf{S}^+ - \mathbf{S}^-$. Retaining participants present on both sides in both matrices preserves catalytic and cofactor requirements. Their net coefficients cancel in $\mathbf{S}$, but they still govern reaction enabling and define distinct reaction complexes (Fig. 2a,b). In Petri-net terms, such shared species form self-loops, and the net incidence matrix alone cannot determine whether transitions are enabled [17]. Construction does not establish elemental balance automatically, so chemical interpretation requires suitable input equations and annotations. The conditions under which a reaction network represents genuine chemical transformations are themselves a subject of study [24]. Assigning known molecules and allowed reaction mechanisms to an abstract network poses a separate chemical realizability problem [25]. At the molecular level, Lewis-labeled graph transformations provide complementary checks of electron availability and conserve represented electrons and formal charge in atom-balanced transformations [26].

Reaction networks can be constructed from curated databases, generative chemistry, or standard model files. The `Query` module parses reaction equations from `KEGG`, `Construct` expands networks from chemical

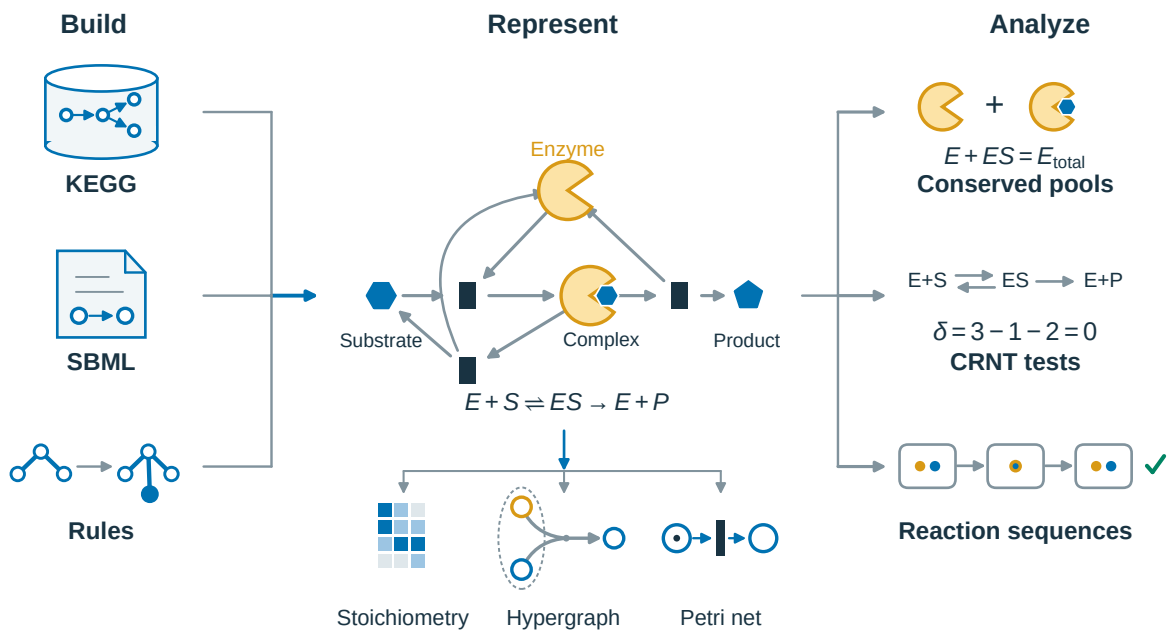


Figure 1 Overview of `SynKit CRN` workflows across construction, representation, and structural analysis. **Build:** Reaction networks imported from curated databases (`KEGG`), exchange files (`SBML`), or generated by rule expansion are mapped to a unified data structure. **Represent:** An illustrative enzyme mechanism ($E + S \rightleftharpoons ES \rightarrow E + P$) is converted into three complementary views: stoichiometric matrices, directed hypergraphs, and Petri nets, ensuring that multi-reactant requirements (E and S) are captured as joint inputs. **Analyze:** These structural representations enable exact conservation analysis (identifying the conserved pool $E + ES$), chemical reaction network deficiency calculation ($\delta = 0$), and discrete pathway realizability through verified firing sequences. Blue circles denote metabolites, orange circles denote enzymes, and dark bars represent reactions.

seeds using reaction rules, and `IO` imports models from `SBML`. During construction, reversible reactions can be partitioned into directed steps. Users can also choose whether to retain cofactors (such as ATP and NADH) and boundary species, which determines the resulting conserved moieties, network deficiency, and discrete pathway realizability.

Some database equations need care before analysis. `KEGG` writes polyphosphate glucokinase (`R02189`) as `C00404 + C00267 <=> C00404 + C00668`, because one identifier stands for both $(\text{Phosphate})_n$ and $(\text{Phosphate})_{n-1}$. Read literally, the polymer looks like a retained catalyst. Chemically, it donates one phosphate. `SynKit` keeps such equations as written, and we exclude them from the participant-retention examples below.

In `SBML`, a catalyst appears either as a species listed on both sides, which `SynKit`'s reader keeps, or as a modifier, which it does not import. Six of the ten `BioModels` entries used here contain modifiers. Adding them to both sides as participants changes the deficiency of four (Table S3). Because modifiers also encode inhibitors and activators, we do not convert them automatically.

2.2 Conserved moiety and stoichiometric analysis

The stoichiometric matrix $\mathbf{S}$ records the net change in every species for each reaction. For concentrations c and reaction rates $v(c)$, the dynamics are written as $\dot{c} = \mathbf{S}v(c)$. A vector y satisfying $y^T\mathbf{S} = 0$ defines a conserved linear quantity $y^T c$, because the reactions can redistribute the participating species without changing this weighted total. `SynKit CRN` computes a rational basis for the left nullspace of $\mathbf{S}$ and converts

its vectors to primitive integer representatives. These vectors describe all rational conservation relations, but they need not be non-negative or identify the smallest conserved pools.

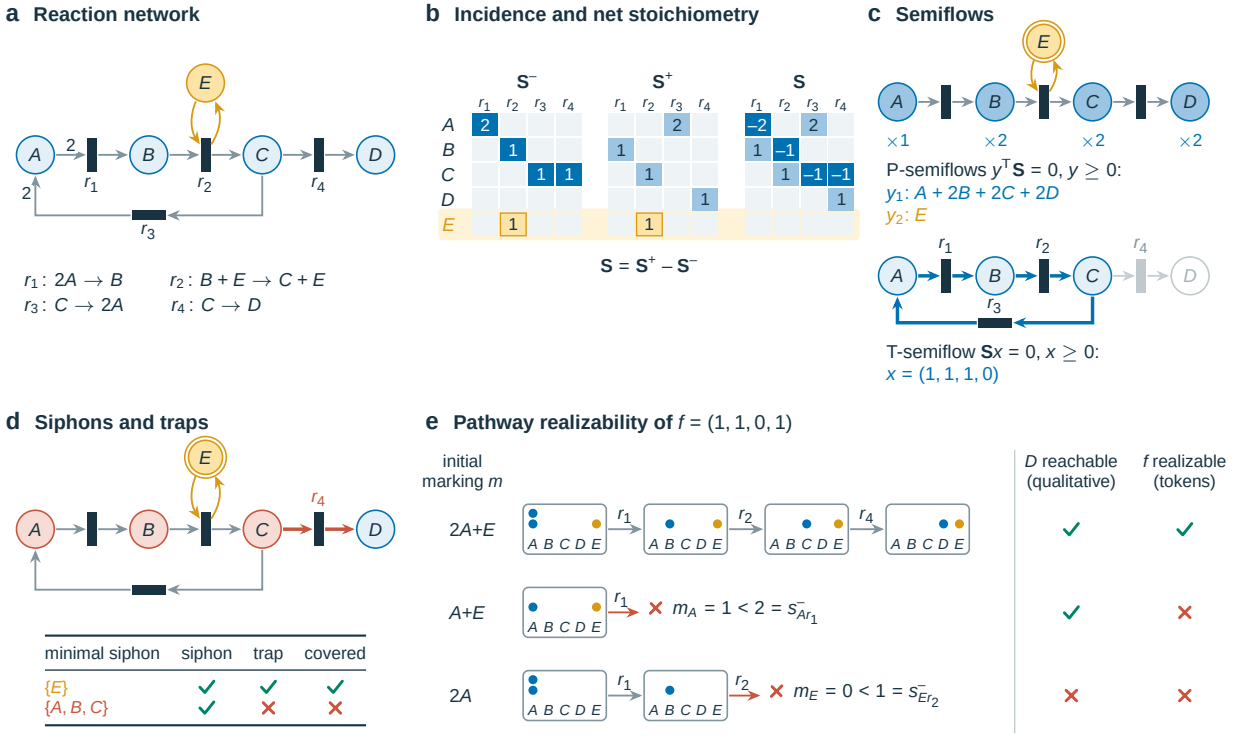


Figure 2 Structural objects on one toy network, colored as in Fig. 1, with red marking blocked steps or uncovered sets. **(a)** Petri net (circles: species, bars: reactions, arc labels: coefficients other than one). The enzyme E is consumed and returned by r_2. **(b)** Reactant, product, and net incidence matrices. The row of E cancels in $\mathbf{S}$, so the net matrix alone does not show that r_2 requires E. **(c)** P-semiflows (shaded, with weights) are conserved pools. The T-semiflow (heavy arcs) is a balanced reaction combination that fixes no firing order. **(d)** Minimal siphons. $\{E\}$ contains a conserved pool and is covered, whereas $\{A, B, C\}$ drains through r_4 (heavy) and is uncovered. **(e)** Firing from three initial markings (dots: tokens). From $A + E$, qualitative reachability reaches D although no reaction is enabled. Inequalities give the failed condition $m_x \geq s^-_{xr}$.

Minimal P-semiflows are computed separately by exact Farkas elimination [27]. They are non-zero, non-negative integer vectors in the left kernel of $\mathbf{S}$ with inclusion-minimal supports, normalized by their greatest common divisor. Their supports identify minimal conserved pools of the represented network [6]. A P-semiflow guarantees mathematical conservation, whereas assigning its support to a chemical moiety requires species annotations and biochemical interpretation.

T-semiflows satisfy $\mathbf{S}x = 0$ with $x \geq 0$ and describe balanced combinations of reactions whose total net species change is zero. They are reaction-space relations, not conserved species pools (Fig. 2c). A T-semiflow fixes neither a firing order nor an initial marking, and it does not imply thermodynamic feasibility. All-positive left and right kernel vectors define conservativity and consistency, respectively. To determine positive-kernel feasibility, a linear program proposes a candidate vector that is accepted only after exact rational verification. Feasible networks are certified by an exact all-positive vector, infeasible networks by a rational separating vector, and unresolved solver outcomes remain undecided. Numerical kernel and Jacobian-pattern checks can also characterize approximate subspaces and qualitative matrix patterns, but because floating-point estimates do not isolate integer minimal supports, numerical rounding cannot substitute for exact conservation analysis.

2.3 Deficiency analysis and multistationarity bounds

Complexes are distinct reactant or product multisets. With n complexes, ℓ connected components of the complex graph (linkage classes), and stoichiometric rank s, deficiency is $\delta = n - \ell - s$. A network is weakly reversible when every linkage class is strongly connected. `SynKit CRN` calculates ranks exactly over $\mathbb{Q}$ and reports the deficiency of the network and of each linkage class, weak reversibility, and the terminal strong linkage classes.

For positive mass-action rate constants, a weakly reversible deficiency-zero network has exactly one positive steady state in each positive stoichiometric compatibility class $(c_0 + \operatorname{im} \mathbf{S}) \cap \mathbb{R}^X_{>0}$, and it is locally asymptotically stable relative to that class [28, 7]. A deficiency-zero network that is not weakly reversible has no positive steady state. The Deficiency One Theorem gives at most one positive steady state per positive stoichiometric compatibility class when each linkage-class deficiency is at most one, their sum equals δ, and each linkage class has one terminal strong linkage class, while weak reversibility additionally ensures existence [29]. Failure of these hypotheses leaves the corresponding theorem undecided. Introductions to this body of theory for biologists are available [8]. Criteria that reach beyond deficiency, including injectivity [30], concordance [31], and graph-theoretic multistationarity tests [32], are not implemented here, and a network left undecided by the deficiency theorems is not thereby shown to be multistationary.

2.4 Pathway realizability and siphon analysis

A siphon is a non-empty species set for which every reaction producing a member also consumes a member. Once empty, it cannot refill under Petri-net firing semantics. A trap has the dual property that every reaction consuming a member also produces a member, so a marked trap cannot become empty [17]. `SynKit CRN` enumerates minimal siphons using implication propagation and closure minimization, within the branch-and-bound approach to structural enumeration [33]. Producing a selected species forces selection of a reactant. Selected and excluded species are propagated before branching. Each reported minimal set excludes its supersets from further search, and a maximal-closure deletion test verifies minimality. Traps use the dual incidence (Fig. 2d). This prunes many subsets but does not remove worst-case combinatorial growth.

The package checks whether every minimal siphon contains the support of a P-semiflow, and we call such a siphon covered. Under the boundedness and kinetic assumptions of the corresponding persistence result, this provides a sufficient persistence certificate for positive initial conditions [34]. An uncovered siphon means the test is insufficient, and does not prove extinction or identify a unique nutrient supplementation strategy. The returned details expose these uncovered sets.

A marking m specifies the non-negative integer inventory of each species. Reaction r is enabled precisely when $m_x \geq s^-_{xr}$ for every species, and firing gives $m' = m + \mathbf{S}_{\cdot r}$. Qualitative reachability instead accumulates reachable species sets and ignores consumption and multiplicity. Given a requested flux $f \in \mathbb{N}_0^{|\mathscr{R}|}$, realizability search seeks an enabled sequence firing each reaction exactly f_r times [9]. A positive result includes a certificate consisting of an ordered reaction list that can be checked by updating the inventory after every

step. An exhaustive negative result is `unrealizable`, while reaching a state or depth limit is `inconclusive`. The minimal-siphon interface defaults to a 200 000-node search budget and raises `SiphonSearchLimit` on incomplete enumeration. Requesting partial output emits a warning. The BioModels experiment uses an explicit 1 000 000-node budget and records completion status. Fig. 2e contrasts the two notions on one requested flux. Certificates establish firing in the specified discrete model only.

2.5 Implementation and software comparison

Users access the software through a Python interface, which calls a C++17 library for computationally intensive tasks. These native routines compute exact rational nullspaces, CRNT deficiency invariants, minimal semiflows, siphon enumeration, and discrete firing sequences. To eliminate floating-point rounding errors and ensure exact certificates, linear-algebraic and combinatorial routines use arbitrary-precision integer arithmetic via Boost.Multiprecision. For generative chemistry, `SynReactor` applies double-pushout graph transformations with implicit hydrogen counts. Its native C++20 controller performs candidate construction, frontier updates, matching, rewriting, and reaction integration from seed molecules. The controller and benchmark scripts are included in the version 1.7.0 release described under Availability of data and materials. Following the `MØD` strategy framework [35], each rule application must use at least one newly generated molecule.

`SynKit` imports static integer reaction topology from `SBML` Levels 2 and 3 and exports Level 3 Version 2 Core. Species identifiers, compartments, and integer stoichiometries are preserved. Fractional coefficients and `stoichiometryMath` are rejected rather than approximated. Reversible reactions expand into directed copies with distinct source identifiers. An omitted `reversible` attribute defaults to true in Level 2. Level 3 requires the attribute: omission raises in strict mode, while the default lenient mode warns and assumes false. Kinetic laws, events, rules, modifiers and package extensions are outside the imported semantics. Exported topology can provide a foundation for subsequent parameterization in kinetic simulators like `Tellurium` or constraint-based packages like `COBRApy`.

Through this design, `SynKit` occupies a distinct niche in the systems biology software ecosystem (Table 1). Established packages typically specialize in continuous kinetic simulation (`Tellurium`), flux-balance analysis (`COBRApy`), or algebraic deficiency testing (`CRNT4SBML`). While `crnpy` evaluates exact semiflows and `MolFoundry` solves rule-based hyperflows, `SynKit` unifies rule generation, deficiency, minimal siphons, and discrete firing certificates on a shared model. Downstream analyses can then build on exported `SBML` networks. Section 1 of the Supplementary Information and Listing S1 provide complete implementation details and usage examples.

2.6 Validation and benchmark design

We validate the toolkit through analytical benchmarks, runtime scaling experiments, and biochemical case studies. First, structural correctness is evaluated across twelve reference networks spanning enzyme kinet-

Table 1 Feature comparison of biochemical reaction-network software (✓ supported, (✓) partial or indirect as detailed in footnotes, × not supported). Capabilities are determined from official documentation and source code. `CRNT4SBML` is experimentally evaluated on shared benchmark networks (Table 2), while formose generation is compared with `MØD` and `MolFoundry`, and fixed-route realizability with `MolFoundry` (Section 3.5). The `SynKit` column describes its chemical reaction network (CRN) submodule.

	`SynKit` (this work)	`CRNT4SBML` [13]	`CoNtRol` [36]	`crnpy` [37]	`COBRApy` [11]	`MØD` [4]	`MolFoundry` [38]	`Tellurium` [14, 15]
Network construction and exchange								
`KEGG` pathway import	✓	×	×	×	×	×	×	×
`SBML` import	(✓)[a]	✓	×	✓	✓	×	×	✓
`SBML` export	(✓)[a]	×	×	✓	✓	×	✓	✓
Rule-based network generation	✓	×	×	×	×	✓	✓	×
Structural analysis								
Deficiency and weak reversibility (CRNT)	✓	✓	✓	✓	×	×	×	×
Conservation laws (left-nullspace basis)	✓	✓	×	✓	×	×	×	✓
Minimal semi-positive conservation laws	✓	×	×	✓	×	×	×	×
Elementary flux modes or T-semiflows	✓	×	×	✓	×	×	×	×
Minimal siphons and traps	✓	×	(✓)[b]	×	×	×	×	×
Dynamics and pathway feasibility								
Multistationarity analysis	×	✓	✓	(✓)[c]	×	×	×	×
Pathway realizability (firing sequence)	✓	×	×	×	×	(✓)[d]	✓[f]	×
Constraint-based modeling (e.g., FBA)	×	×	×	×	✓	×	(✓)[f]	×
Kinetic simulation	×	(✓)[e]	×	×	×	×	✓[g]	✓

[a] Fixed-integer reaction structure: Level 2/3 import and Level 3 Version 2 export, excluding kinetic laws, rules and modifiers. Level 2 omitted reversibility flags default to true; Level 3 flag handling is described in Implementation.
[b] Siphons underlie persistence tests, but minimal-siphon enumeration is not exposed as an output.
[c] `crnpy` supplies Gröbner-basis, elementary-mode, and ACR machinery rather than bistability search.
[d] `MØD` evaluates integer hyperflows rather than discrete firing certificates over a finite marking.
[e] ODE simulation is provided in support of `CRNT4SBML`'s bistability analysis.
[f] `MolFoundry` solves integer hyperflows over derivation graphs (Gurobi or SMT) for firing orders.
[g] Rule-based stochastic simulation with on-demand network expansion.

ics, phosphorylation cycles, bistable switches, and motifs exhibiting concentration robustness, with shared CRNT invariants independently confirmed by `CRNT4SBML`. Second, computational scaling for exact siphon and conserved-pool enumeration is profiled across synthetic topologies to establish practical tractability limits. Third, `SBML` model exchange and deficiency preservation are tested across ten reaction-bearing `BioModels` entries, up to 504 species and 1 654 directed reactions, plus one reaction-free qualitative control.

Two mechanistic case studies test execution of prescribed reaction multisets from finite metabolite pools. In `KEGG` glycolysis, we assess how retaining versus omitting currency metabolites affects a fixed flux from an initial inventory containing only glucose. In rule-generated formose chemistry, we compare qualitative hypergraph reachability with discrete firing of two selected routes under finite precursor budgets. Each positive decision has an ordered reaction sequence, and each negative decision follows exhaustive search over the selected multiset and initial marking.

3 Results and discussion

3.1 Reference-network validation and comparison with `CRNT4SBML`

All twelve reference networks match their analytical and algebraic ground-truth values (Table 2, with sources and validation objectives in Table S1). The benchmark suite spans classic enzyme kinetics (Michaelis–Menten and its reversible counterpart), multisite phosphorylation cycles [39], the bistable Edelstein switch [40], and the Shinar–Feinberg robustness motif [41]. Constructed synthetic cases also verify boundary conditions. Disconnected reversible pairs confirm that independent conserved pools are correctly isolated without conflating disjoint subnetworks, while the open reactor ($0 \to A \to B \to 0$) tests exact handling of inflow and outflow through the zero complex.

Table 2 Structural invariants of the twelve reference networks: species, directed reactions, complexes (n), linkage classes (ℓ), stoichiometric rank (s), deficiency (δ), conservation dimension (laws), and weak reversibility (WR). Every network matches its analytical and algebraic ground truth, and `CRNT4SBML` reproduces all eight shared structural quantities.

Network	Species	Reactions	n	ℓ	s	δ	Laws	WR
Reversible isomerization	2	2	2	1	1	0	1	yes
Irreversible isomerization	2	1	2	1	1	0	1	no
Isomerization cycle	3	3	3	1	2	0	1	yes
Disconnected reversible pairs	6	6	6	3	3	0	3	yes
Pairs with stoichiometry	6	6	6	3	3	0	3	yes
Michaelis–Menten	4	3	3	1	2	0	2	no
Reversible Michaelis–Menten	4	4	3	1	2	0	2	yes
Edelstein switch	3	6	5	2	2	1	1	yes
Phosphorylation: one site	6	6	6	2	3	1	3	no
Phosphorylation: two sites	9	12	10	2	6	2	3	no
Shinar–Feinberg motif	2	2	4	2	1	1	1	no
Open reactor	2	3	3	1	2	0	0	yes

To benchmark structural interoperability, `CRNT4SBML` 0.0.15 independently parsed all twelve exported `SBML` models. Across all twelve benchmarks, both tools yielded identical results for all eight shared structural quantities (Table 2), confirming topological consistency across independent implementations. As a separate numerical audit, SVD nullspace bases have residuals between zero and 5.3×10^{-16}. Multiplying those bases by 10^3 and rounding to integers yields invalid conservation vectors on three of the twelve models (Table S1). These failures concern that specific reconstruction heuristic. Exact rational elimination returns integer representatives satisfying $y^T \mathbf{S} = 0$ in every case, with their conservation dimensions verified against the analytical benchmarks.

3.2 Computational cost of exact structural enumeration

Enumerating minimal siphons (Fig. 2d) and exact conserved pools (Fig. 2c) identifies species sets that stay empty once depleted and mass conservation laws, but naive search scales exponentially with $2^N - 1$ candidate subsets. To prevent combinatorial explosion, implication pruning propagates stoichiometric constraints along active reactions during search (Fig. 3a). In reversible chains, selecting an initial species forces each adjacent neighbor, isolating the entire chain as the unique minimal siphon without evaluating

spurious sub-combinations. Across five repeats per size, pruned search returns identical siphon sets with a 41.6-fold speedup at 21 species (0.89 ms versus 37.11 ms for exhaustive search, Fig. 3b).

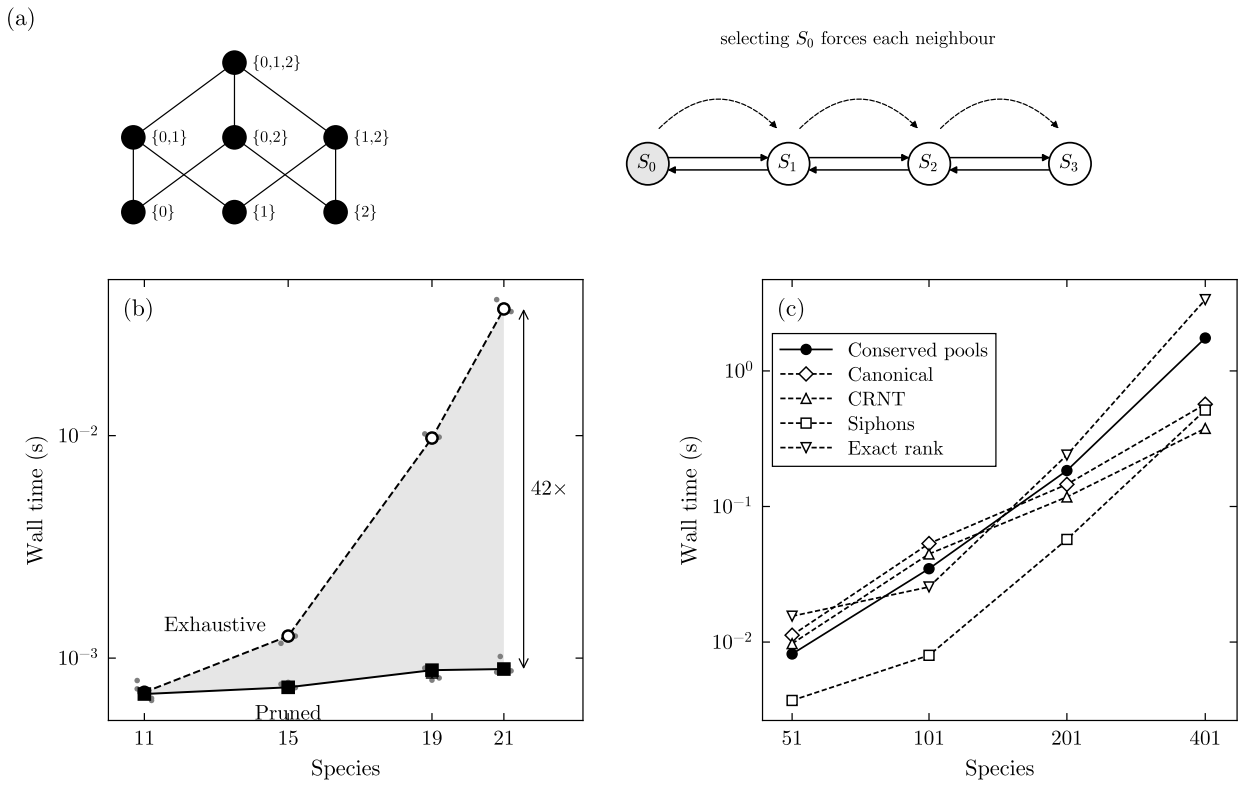


Figure 3 Search algorithms and runtime scaling on reversible-chain networks. **(a)** Schematic of exhaustive power-set search versus implication pruning, where selecting S_0 forces all chain species into the minimal siphon. **(b)** Minimal-siphon runtimes across five repeats per size (points: individual runs, markers: medians and IQRs, bracket: speedup ratio at 21 species). **(c)** Wall times for five structural analyses across chains up to 401 species.

The broader benchmark suite evaluates scalability across directed chains, reversible chains, and random sparse bimolecular networks (Fig. S3). All 60 requested configurations complete without numerical failure. On the 401-species reversible chain, minimal siphons (Fig. 2d) take 0.515 s, canonicalization 0.568 s, exact conserved pools (Fig. 2c) 1.75 s, and standalone exact rank 3.36 s (Fig. 3c). The random networks with 49, 99, 197, and 395 species contain 331, 272, 47, and 87 minimal siphons, completing siphon and conserved-pool enumeration in 0.149 s and 0.0697 s at 395 species. Stoichiometric implication pruning and rational nullspace methods thus keep exact structural enumeration computationally practical on networks with hundreds of species.

3.3 Structural `SBML` round-trip validation in `BioModels`

To test real-world structural interoperability and scalability, we evaluated eleven curated `BioModels` [42] entries spanning diverse sizes, including ten reaction networks and one qualitative network control (`SBML-qual`). All ten reaction-bearing models round-trip through the `SBML` reader and writer without loss of stoichiometry, and their ranks and deficiencies agree with an independent exact-rational computation from the `SBML` reaction lists (Table S2). Even on the largest model (`BIOMD0000000255`, with 504 species and 1 654 directed reactions), `SynKit` identifies 23 minimal siphons in 1.2 s and 17 exact conserved pools in 9.74 s (Fig. 4a). Test-

ing siphon coverage by conserved pools gives a sufficient condition for structural protection from extinction. In four of the ten models (BIOMD0000000064, BIOMD0000000019, BIOMD0000000175 and BIOMD0000000255), at least one minimal siphon contains no conserved-pool support, so the sufficient persistence condition [34] cannot certify these models (Table S2). This does not imply that any species goes extinct. It identifies the species sets that a kinetic analysis should examine first. The qualitative control BIOMD0000000562 is correctly identified as reaction-free for Core structural analysis.

Participant retention matters in curated models as well. In BIOMD0000000404, two dephosphorylation reactions of phosphorylated CheY (Yp) list their catalysts, the phosphatase CheZ (Z) and the receptor complex Tasp_WA, as both reactant and product, for example $\mathtt{Yp} + \mathtt{Z} \rightarrow \mathtt{Y} + \mathtt{Z}$. Cancelling these participants, as net-stoichiometry tools do, leaves $\mathbf{S}$ unchanged but lowers the deficiency from 13 to 11 (Table S3), and it lets dephosphorylation fire without CheZ.

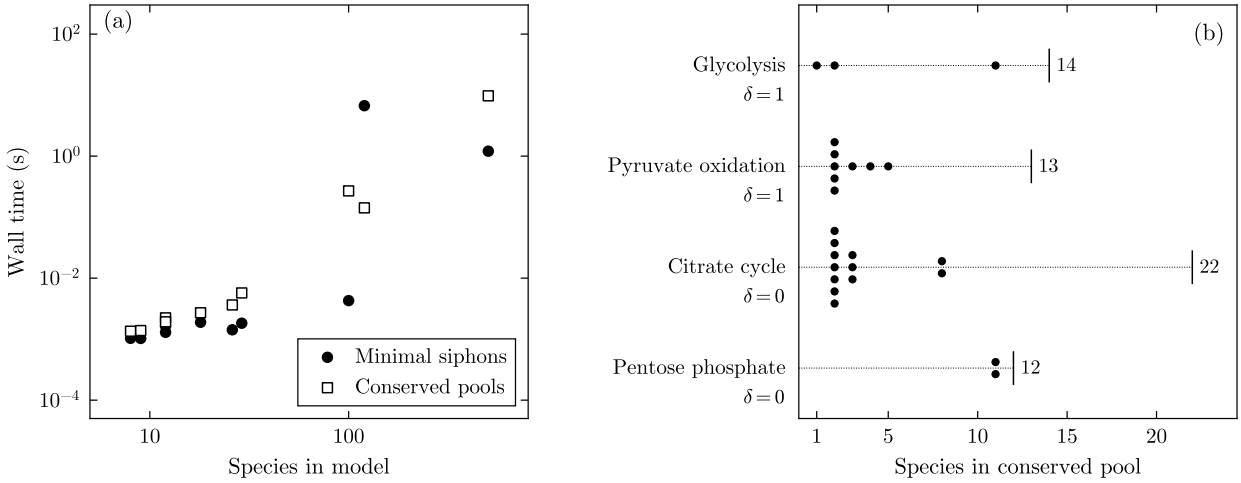


Figure 4 Enumeration runtimes in BioModels and conserved pools in KEGG modules. **(a)** Wall times for minimal-siphon and exact conserved-pool enumeration in ten reaction-bearing BioModels models. All ten models complete both analyses under the stated limits. **(b)** Minimal non-negative conserved pools in four reduced KEGG modules. Each dot is one pool placed at its number of species, pools of equal size are stacked, and the dotted line spans the module species, ending at the module size.

3.4 Conservation and cofactor requirements in glycolysis

Four KEGG [1] pathway modules illustrate structural conservation across metabolic subsystems: glycolysis (M00001), pyruvate oxidation (M00307), the citrate cycle (M00009), and the pentose phosphate pathway (M00004). Their conserved-pool supports appear in Fig. 4b. In the reduced citrate-cycle network, minimal non-negative conservation vectors automatically isolate key metabolic moiety pools, recovering nucleotide pairs, a quinone pair, a ferredoxin pair, coenzyme A, and a lipoyl cofactor pool without manual annotation.

The reduced glycolysis model contains 14 species and 15 directed reactions ($n = 16$, $\ell = 4$, rank $s = 11$, and deficiency $\delta = 1$). Its primary conserved moiety spans eleven carbohydrate intermediates, assigning weight two to hexoses and one to trioses (Fig. 5), capturing exact three-carbon backbone conservation across aldolase cleavage. Glucose and oxidized ferredoxin form uncovered singleton siphons: if either species is absent, the reduced network cannot produce it. This property alone does not require supplementation for every flux. The prescribed glucose-to-pyruvate flux below does not use the ferredoxin bypass.

From a single initial glucose molecule, the prescribed glycolytic flux yields a certified 15-step pathway realization terminating in two pyruvates, exploring 30 intermediate markings. Aldolase cleavage generates two triose equivalents that feed the repeated lower payoff phase. Step-by-step certificate replay confirms intermediate availability at every enzyme step and verifies exact invariance of the weighted three-carbon backbone pool ($2 \times$ hexoses $+ 1 \times$ trioses $= 2$). Attempting to execute the lower payoff phase from glucose alone fails because the preparatory intermediates have not yet formed.

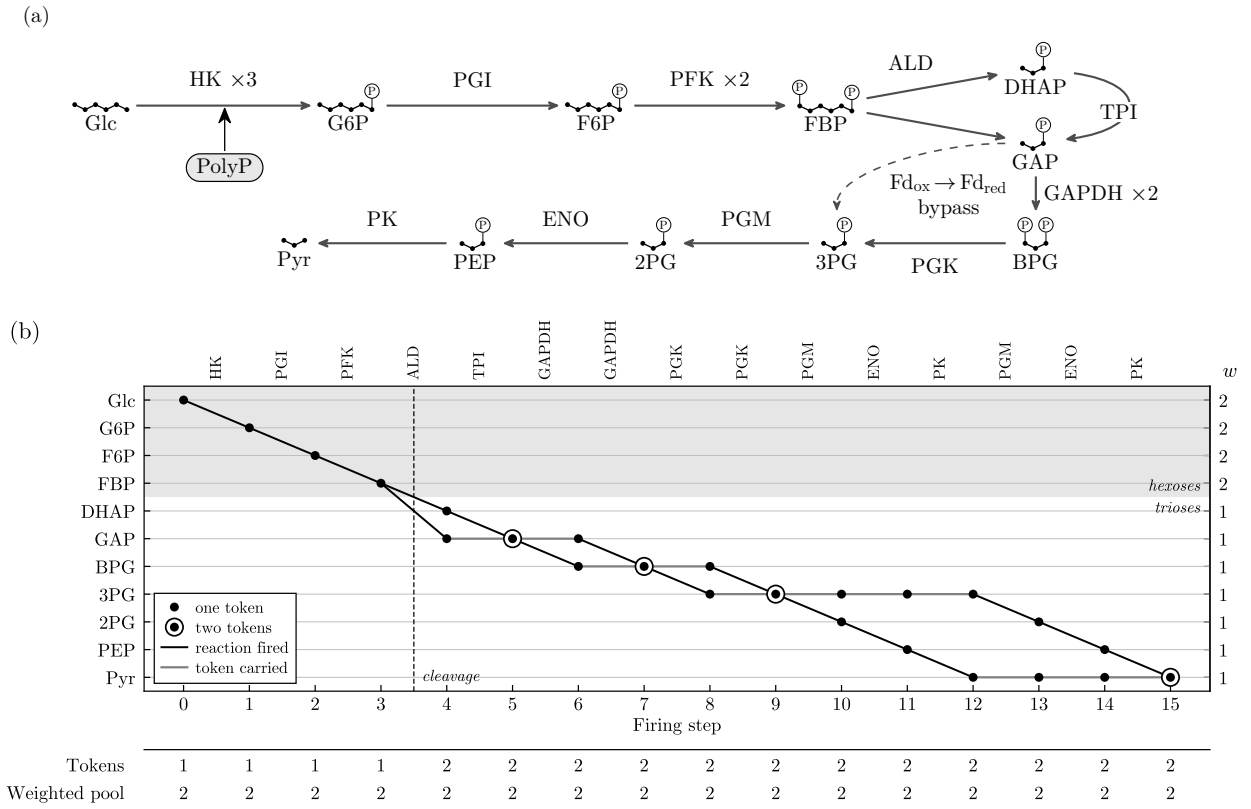


Figure 5 Conserved pools and pathway realizability in reduced KEGG glycolysis (M00001). **(a)** Pathway schematic showing the KEGG polyphosphate entry (R02189), kept as written (Section 2.1), and the ferredoxin bypass (dots: carbons, circled P: phosphate). **(b)** Token markings along the certified 15-step conversion of one glucose to two pyruvates (single dot: one token, ringed dot: two tokens). Black lines trace consumed intermediates, gray lines carry unchanged pools, column headers name enzymes, and the dashed line marks aldolase cleavage. Shaded rows indicate hexoses, with pool weights w on the right. Bottom rows trace total carbohydrate tokens and the invariant weighted pool ($2 \times$ hexoses $+ 1 \times$ trioses $= 2$).

Restoring currency metabolites yields the 24-species model. From one glucose and zero tokens of every other species, the prescribed flux cannot fire: ATP and ADP form an initially empty siphon in the selected reaction subnet, and hexokinase lacks ATP (Fig. 6). Supplying ATP alone still leaves net deficits of ADP, NAD^+ and orthophosphate. A stronger control supplies two ADP, two NAD^+ and two orthophosphate tokens alongside glucose. With one initial ATP, $m_0 + \mathbf{S}f \geq 0$, yet the flux remains unrealizable because its two ATP-consuming preparatory steps precede ATP production. Raising the initial ATP count to two yields an independently replayed 15-step certificate ending with two pyruvates and four ATP. These decisions concern the same prescribed multiset under different finite inventories (Table S7).

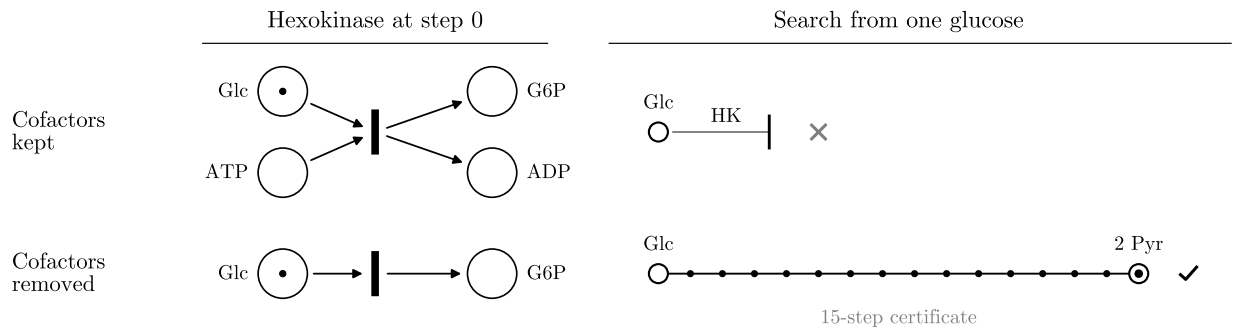


Figure 6 Pathway realizability of the prescribed glycolytic flux from one glucose under retained versus removed currency metabolites. Left: hexokinase (`R01786`) at step 0. Right: pathway reachability from initial glucose (open circle) through intermediate steps (dots) to two pyruvates (ringed dot).

3.5 Generative chemistry: carbon conservation and precursor requirements in formose

The formose reaction, the classic prebiotic condensation of formaldehyde into diverse sugars [21], provides an open-ended, rule-generated benchmark complementary to curated metabolic networks. Seeded with formaldehyde (C_1) and glycolaldehyde (C_2), four core transformation rules (aldol addition, retro-aldol cleavage, and reversible keto-enol tautomerization through an enediol intermediate) iteratively construct the reaction space. After four expansion rounds, the network comprises 37 species and 44 directed reactions ($n = 54$ complexes, $\ell = 17$ linkage classes, rank $s = 36$, and deficiency $\delta = 1$), reaching decose (C_{10}) species. Median cumulative generation time with the native `SynReactor` DPO controller is 0.0283 s across 20 timed runs after three warm-ups.

Matched expansions in `MØD` 1.0.0.75 [4] and `MolFoundry` 0.1.11 [38] independently validate network generation across six expansion depths (Fig. 7a). At depth six, all three tools generate identical sets of 10 572 species and 11 239 stoichiometric reactions. Median cumulative generation times with all tool processes bound to one logical CPU are 1.86 s for native `SynReactor` DPO, 2.22 s for `MØD`, and 15.1 s for `MolFoundry` (Fig. 7b and Table S8). These end-to-end API measurements encompass rule parsing, frontier updates, and Python graph construction for `SynKit` without isolating native kernel execution.

Exact nullspace analysis of the four-iteration stoichiometric matrix reveals a one-dimensional left nullspace. Its positive primitive conservation vector agrees identically with the heavy-atom carbon count of every generated molecule, confirming that automated graph transformation preserves carbon conservation across all 37 species (Fig. 8b).

Qualitative hypergraph search reaches representative aldotetrose (T_4) and aldohexose (T_6) targets from the seed species. We then test two prescribed routes: the tetrose route consumes two glycolaldehyde molecules and the hexose route three, while neither consumes formaldehyde. Consequently, six-carbon inventories containing one (M_3), two (M_4), or three (M_5) glycolaldehydes support neither route, only the tetrose route, or both routes, respectively (Fig. 8c). These decisions establish the precursor requirements of the selected reaction multisets. They do not rule out other routes from the mixed inventories. Separately, an eight-species

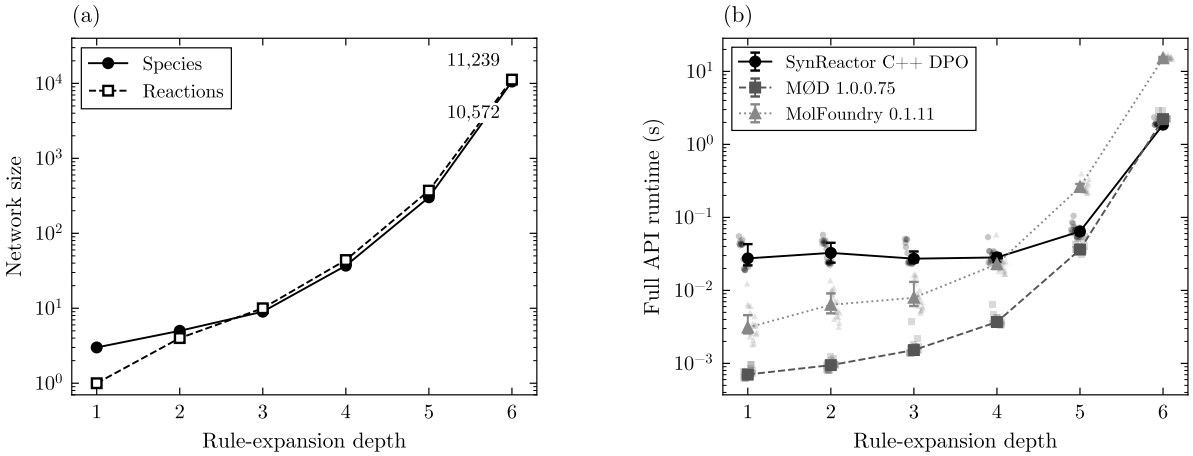


Figure 7 Formose network growth and full API runtime. **(a)** Species and directed stoichiometric reactions across six rule-expansion depths, with all three tools generating identical sets. **(b)** Full API runtime of native `SynReactor` DPO, `MØD`, and `MolFoundry`. Faint points show 20 timed runs per tool and depth after three warm-ups, large markers show medians, and error bars span the first and third quartiles. Points are offset horizontally for visibility, and times include per-call rule and seed preparation and network construction without imports or engine warm-up. Both panels use logarithmic vertical axes.

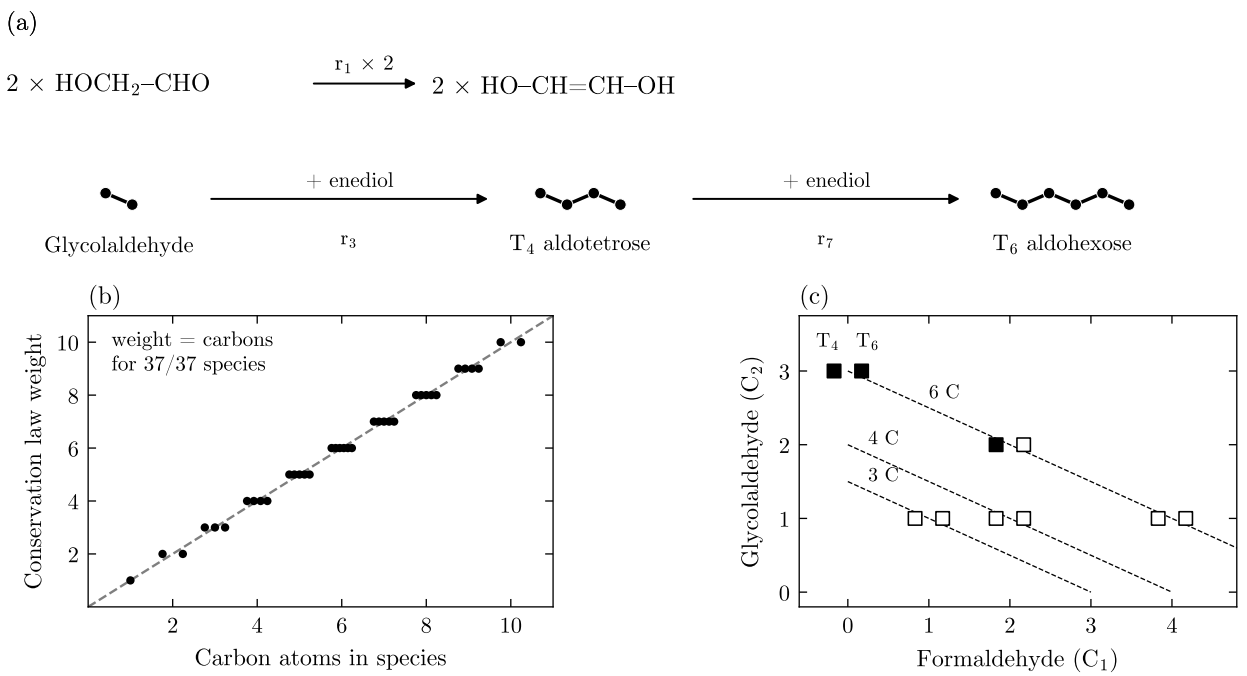


Figure 8 Carbon conservation and realizability of two prescribed routes in the generated formose network. **(a)** Target hexose route requiring three glycolaldehyde precursors via two enolizations (r_1) and two aldol additions (r_3, r_7), each addition consuming one enediol. **(b)** Left-nullspace conservation weights against carbon atom counts across all 37 species (points jittered horizontally, dashed line: $y = x$). **(c)** Five initial inventories by formaldehyde and glycolaldehyde count, with dashed lines joining inventories of equal total carbon. The left and right squares of each inventory refer to the tetrose (T_4) and hexose (T_6) routes, filled for a verified certificate and open for an exhaustive negative decision for that multiset. Inventories with six carbon atoms differ in which routes they can supply.

minimal siphon contains glycolaldehyde, its enediol, and both targets but excludes formaldehyde. If initially empty, this set stays empty, showing why formaldehyde alone cannot initiate target production within the generated network. All ten selected-route decisions match independent evaluation in `MolFoundry`, and all positive certificates pass independent replay (Supplementary Section S6).

4 Conclusions

The CRN submodule of `SynKit` unites stoichiometric conservation, chemical reaction network theory, and pathway realizability on a bipartite representation preserving distinct reactant and product incidences. Across curated models, metabolic modules and generative chemistries, these analyses demonstrate that net stoichiometric reduction obscures essential physical constraints. Retaining shared participants preserves network deficiency and keeps catalysts such as the phosphatase CheZ required for firing. Similarly, omitting currency metabolites exposes carbon backbone realizability, while their retention makes execution depend on the specified cofactor and phosphate inventories. In the generated formose network, carbon conservation alone does not establish execution of the selected routes, whose reaction multisets require particular glycolaldehyde inventories.

These analyses characterize structural enabling and discrete token sufficiency rather than continuous reaction rates, concentrations, thermodynamic driving forces, or *in vivo* regulatory kinetics. Reachability certificates evaluate prescribed reaction multisets rather than unrestricted *de novo* route synthesis, and exact siphon and semiflow enumeration faces combinatorial scaling on dense networks. Structural `SBML` exchange is confined to static integer reaction topology, excluding kinetic rate laws, rules, modifiers and non-core extensions.

Future work will use estimated reaction free energies ($\Delta_r G$) to filter generated pathways before kinetic simulation. Coupling the incidence representation with constraint-based flux analysis and ODE simulators will connect structural feasibility to quantitative metabolic phenotypes. Reducing cross-boundary serialization will support larger prebiotic reaction networks.

List of abbreviations

ACR: Absolute concentration robustness, **CRN**: Chemical reaction network, **CRNT**: Chemical reaction network theory, **DPO**: Double pushout, **EMP**: Embden–Meyerhof–Parnas, **FBA**: Flux balance analysis, **KEGG**: Kyoto Encyclopedia of Genes and Genomes, **SBML**: Systems Biology Markup Language, **SMARTS**: SMILES arbitrary target specification, **TCA**: Tricarboxylic acid, **WR**: Weakly reversible.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The CRN submodule is distributed with `SynKit` under the MIT License. The source code is available at github.com/TieuLongPhan/SynKit, and the experiments at `Experiment/CRN`. This study uses version 1.7.0 [43]. `KEGG` [1] pathways remain subject to `KEGG` terms, and `BioModels` entries are distributed under the CC0 1.0 Public Domain Dedication.

Use of generative AI

Claude Opus (Anthropic) was used for language editing and TikZ figure formatting. The authors reviewed, verified, and take full responsibility for all manuscript content.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

Author contributions are reported using the CRediT taxonomy. **T.-A.P.**: Software, Validation, Writing (original draft), Writing (review and editing). **T.-L.P.**: Conceptualization, Methodology, Software, Validation, Writing (original draft), Writing (review and editing). Both authors read and approved the final manuscript.

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